

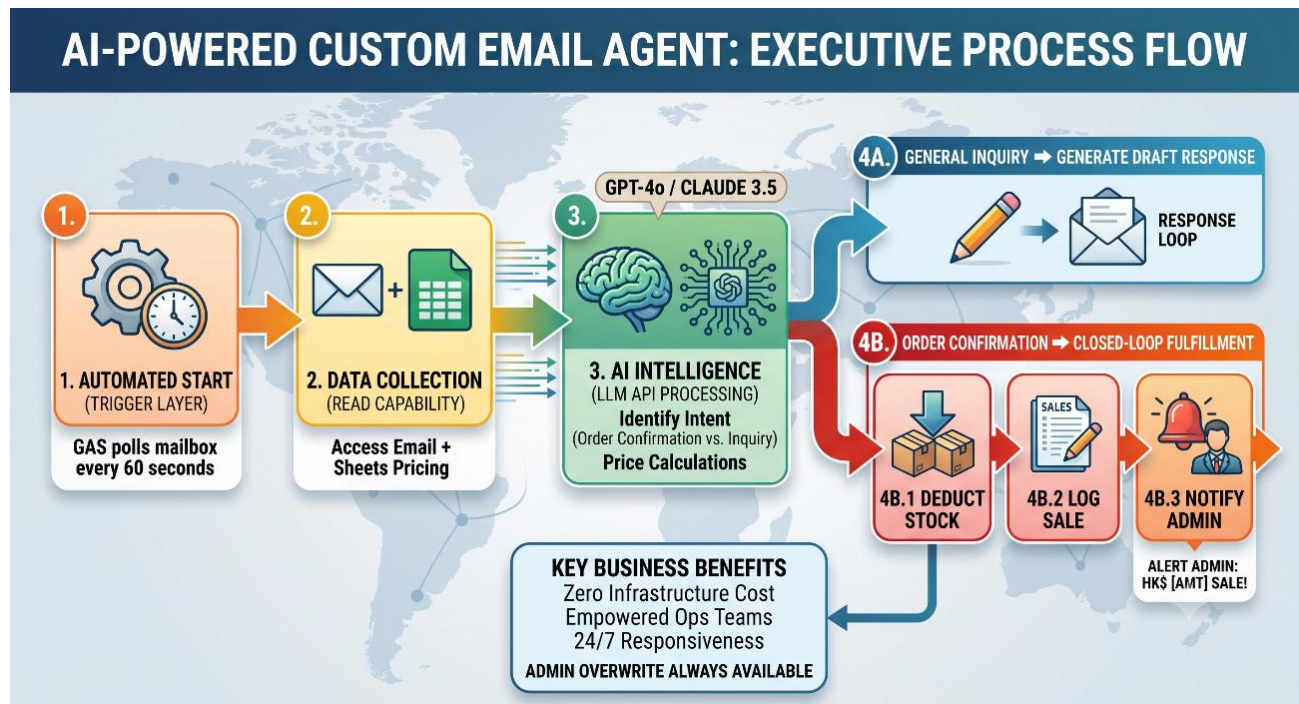
AI-Powered Custom Email Agent

Name: Zhang Guoyong

Executive Summary:

Automated Luxury Fruit Agent Quotation System – Link: [Gerald Studio](#)

This cloud-based project simulates a fruit store's customer service staff, providing 24/7 support, including product introductions, a pricing system, reading and modifying real-time product prices and inventory, and notifying the administrator.



Introduction & Problem Statement:

The project uses AI references.

Project Background: In daily operations, we receive inquiries from customers speaking different languages. Manually replying to emails is time-consuming and laborious, price calculations are cumbersome, and manually updating inventory is prone to errors and untimely.

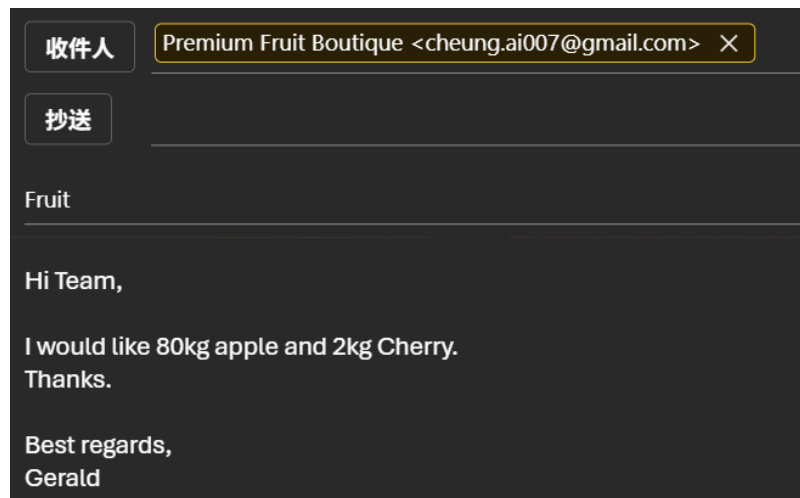
Objectives: To build a 24/7 automated email and warehouse management system to improve response speed.

System Architecture & Technical Implementation:

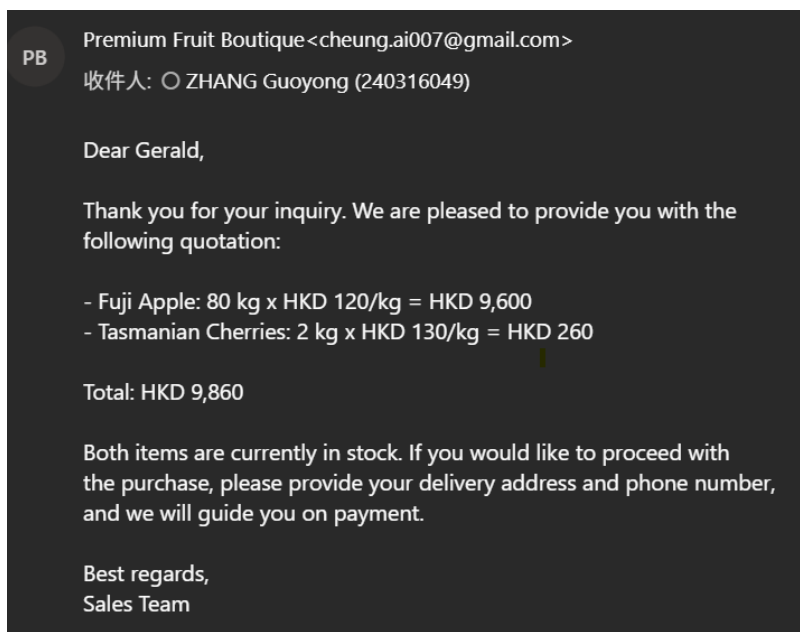
The project is built upon a lightweight, serverless, and highly agile architecture that seamlessly bridges the Google Workspace ecosystem with cutting-edge artificial intelligence. Rather than relying on traditional, costly server hosting, it manages data flow and automation. The system can also be deployed via local LLMs, ensuring absolute data privacy and delivering lower latency.

Core Workflow & Functional Highlights:

- User inquiry & Triggering inventory warning



- AI agent calculates order prices and guides purchases.



- Address and phone number trigger order



● Automatically adjust inventory (Before – Execution – Completed)

A	B	C	D	E	F
FRUIT_NAME_COL	PRICE_COL	Description	INVENTORY_COL	Safety_Stock_COL	Status_Alert
Fuji Apple	120	Per kg	100	30	● Stock is plentiful
Taiwan Mango	180	Per kg - Irwin varie	100	30	● Stock is plentiful
Gold Kiwi	250	Per box - 6 pieces	100	30	● Stock is plentiful
Thailand Durian	350	Per kg - Monthong	100	30	● Stock is plentiful
Organic Banana	80	Per kg - Locally gro	100	30	● Stock is plentiful
Tasmanian Cherries	130	Per kg	100	30	● Stock is plentiful

Execution log		
12:55:09 PM	Notice	Execution started
12:55:10 PM	Info	Processing an email from the <240316049@stu.vtc.edu.hk> "ZHANG Guoyong (240316049)": Fruit
12:55:14 PM	Info	成功回覆郵件給: "ZHANG Guoyong (240316049)" <240316049@stu.vtc.edu.hk>
12:55:15 PM	Info	✓ 成功寫入 Sales_Log: Fuji Apple x 80
12:55:16 PM	Info	Inventory has been updated and administrator has been notified: Fuji Apple, amount: 9600
12:55:16 PM	Info	✓ 成功寫入 Sales_Log: Tasmanian Cherries x 2
12:55:17 PM	Info	Inventory has been updated and administrator: Tasmanian Cherries, amount: 260
12:55:22 PM	Notice	Execution completed

A	B	C	D	E	F
FRUIT_NAME_COL	PRICE_COL	Description	INVENTORY_COL	Safety_Stock_COL	Status_Alert
Fuji Apple	120	Per kg	20	30	● Inventory low
Taiwan Mango	180	Per kg - Irwin varie	100	30	● Stock is plentiful
Gold Kiwi	250	Per box - 6 pieces	100	30	● Stock is plentiful
Thailand Durian	350	Per kg - Monthong	100	30	● Stock is plentiful
Organic Banana	80	Per kg - Locally gro	100	30	● Stock is plentiful
Tasmanian Cherries	130	Per kg	98	30	● Stock is plentiful

● Record orders

	A	B	C	D	E	F	G
1	Order_Date	Customer_Email	FRUIT_NAME_COL	Quantity	PRICE_COL	Total_Amount	Payment_Status
2	2026-07-02 12:55:14	<240316049@stu.vtc.edu.hk>	Fuji Apple	80	120	9600	Pending
3	2026-07-02 12:55:16	<240316049@stu.vtc.edu.hk>	Tasmanian Cherries	2	130	260	Pending

● Alert administrator and inventory warning



cheung.ai007@gmail.com
to me

It looks like this message is in Chinese (Traditional)
[Translate to English](#)



今日銷售與庫存總覽

管理員您好，客戶 "ZHANG Guoyong (240316049)" <240316049@stu.vtc.edu.hk> 剛剛購買了 Tasmanian Cherries (2 件)。

👉 本次銷售金額: HK\$ 260

以下是目前的最新庫存狀態:

水果品項	剩餘庫存	狀態警示
Fuji Apple	20	● 庫存偏低
Taiwan Mango	100	● 庫存充足
Gold Kiwi	100	● 庫存充足
Thailand Durian	100	● 庫存充足
Organic Banana	100	● 庫存充足
Tasmanian Cherries	98	● 庫存充足

Discussions:

This system effectively demonstrates the potential of a serverless architecture. This cost-efficiency makes it an ideal Minimum Viable Product (MVP) for a startup.

Commercial Value and Warehouse Management Integration

This project is designed not only as an intelligent communication tool but as an agile micro-Order Management System (OMS) that bridges the gap between front-end sales and back-end warehouse operations. The system leverages AI to automate the entire fulfillment lifecycle, providing significant operational efficiencies and scalability for commercial applications.

1. Multilingual AI Agent and Standardized Order SOP

The system acts as a 24/7 autonomous digital operator capable of comprehending and processing unstructured email inquiries in any language. By establishing a strict Standard Operating Procedure (SOP), the AI intelligently distinguishes between casual inquiries and definitive purchase intents. Once a customer confirms an order and provides their delivery details, the system automatically validates the request, finalizes the transaction, and issues standardized payment instructions. This capability drastically reduces order processing lead times and minimizes the overhead associated with manual customer service.

2. Order-Driven Real-Time Inventory Synchronization

To address the critical delay between front-end order reception and back-end inventory updates, the AI agent seamlessly integrates with the database to maintain real-time operational accuracy. Upon successfully confirming an order, the system automatically records the transaction in the sales log and instantly deducts the precise quantity of the purchased items from the active inventory. This immediate data synchronization eliminates manual data entry errors, mitigates the risk of overselling, and ensures absolute consistency between the sales front and the warehouse floor.

3. Future Scalability: Secure Advanced WMS via Self-Hosted Local LLMs

While the current cloud-based setup efficiently handles baseline order automation, the architecture is designed for a strategic upgrade to a self-hosted platform utilizing locally deployed Large Language Models (LLMs). This on-premise transition ensures absolute data security and privacy compliance by keeping all sensitive commercial and customer information strictly within the internal network. Furthermore, a self-hosted infrastructure grants unprecedented operational flexibility, allowing the system to rapidly adapt to dynamic business logic and scaling requirements without vendor lock-in. In this advanced phase, the system will migrate from cloud-based spreadsheets to a robust local database format. The AI Agent will act as the central orchestrator, directly querying and operating the local database to execute complex Warehouse Management System (WMS) tasks. By granting the Agent direct database control, it can enforce precise First-In, First-Out (FIFO) batch control for perishable goods and autonomously calculate dynamic pick path optimizations, maximizing warehouse routing efficiency and overall picking productivity.